

MAT2003 - Applied Numerical Methods

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Module-IV

Numerical Differentiation and Integration

Example 4.1.1

Given a cubic polynomial with following data points

x	0	1	2	3
$f(x)$	5	6	3	8

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 0$.

The forward difference table is:

x	y	Δ	Δ^2	Δ^3
0	5			
		1		
1	6		-4	
		-3		12
2	3		8	
		5		
3	8			

To find the derivative at $x = 0$, taking $x_0 = 0$ and applying the relation:

$$\left. \frac{dy}{dx} \right]_{x=x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{\Delta^2 y_0}{2} + \frac{\Delta^3 y_0}{3} - \frac{\Delta^4 y_0}{4} + \dots \right] \quad (1)$$

From table $h = 1$,

Substituting forward difference table values in equation (1), we get

$$\left. \frac{dy}{dx} \right]_{x=0} = \frac{1}{1} \left[1 - \frac{(-4)}{2} + \frac{12}{3} \right] = 7$$

Also,

$$\left. \frac{d^2 y}{dx^2} \right]_{x=x_0} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 + \dots \right] \quad (2)$$

$$\left. \frac{d^2 y}{dx^2} \right]_{x=0} = \frac{1}{1^2} [-4 - 12 + 0] = -16$$

Example 4.1.2

Given a polynomial with following data points:

x	1.0	1.1	1.2	1.3	1.4	1.5	1.6
$f(x)$	7.989	8.403	8.781	9.129	9.451	9.750	10.031

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 1.1$ and $x = 1.5$.

x	$f(x)$	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
1.0	7.989						
		0.414					
1.1	8.403		-0.036				
		0.378		0.006			
1.2	8.781		-0.030		-0.002		
		0.358		0.004		0.001	
1.3	9.129		-0.026		-0.001		0.002
		0.322		0.003		0.003	
1.4	9.451		-0.023		0.002		
		0.299		0.005			
1.5	9.750		-0.018				
		0.281					
1.6	10.031						

To find the derivative at $x = 1.1$, taking $x_0 = 1.1$ and applying the relation:

$$\left. \frac{dy}{dx} \right]_{x=x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{\Delta^2 y_0}{2} + \frac{\Delta^3 y_0}{3} - \frac{\Delta^4 y_0}{4} + \dots \right] \quad (3)$$

From given data $h = 0.1$.

Substituting forward difference table value in equation (3), we get

$$\begin{aligned} \left. \frac{dy}{dx} \right]_{x=x_0} &= \frac{1}{0.1} \left[0.378 - \frac{(-0.030)}{2} + \frac{0.004}{3} - \frac{(-0.001)}{4} + \frac{0.003}{5} - \dots \right] \\ &= 3.9518. \end{aligned}$$

Also,

$$\begin{aligned} \left. \frac{d^2 y}{dx^2} \right]_{x=x_0} &= \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 + \dots \right] \\ \left. \frac{d^2 y}{dx^2} \right]_{x=1.1} &= \frac{1}{(0.1)^2} \left[-0.03 - 0.004 + \frac{11}{12}(-0.001) - \frac{5}{6}(0.003) \right] = -3.74 \end{aligned}$$

To find the derivative at $x = 1.5$, taking $x_n = 1.5$ and applying the relation:

$$\left. \frac{dy}{dx} \right|_{x=x_n} = \frac{1}{h} \left[\nabla y_n + \frac{\nabla^2 y_n}{2} + \frac{\nabla^3 y_n}{3} + \frac{\nabla^4 y_n}{4} + \dots \right] \quad (4)$$

From given data $h = 0.1$.

Substituting the forward difference table value in equation (4), we get

$$\left. \frac{dy}{dx} \right|_{x=1.5} = \frac{1}{0.1} \left[0.299 + \frac{(-0.023)}{2} + \frac{0.003}{3} + \frac{(-0.001)}{4} + \frac{0.001}{5} \right] = 2.8845$$

Also,

$$\left. \frac{d^2y}{dx^2} \right|_{x=x_n} = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n + \dots \right] \quad (5)$$

$$\left. \frac{d^2y}{dx^2} \right|_{x=1.5} = \frac{1}{(0.1)^2} \left[-0.023 + 0.003 + \frac{11}{12}(-0.001) + \frac{5}{6}(0.001) \right] = -2.0083$$

Trapezoidal rule

Trapezoidal rule to evaluate $\int_a^b f(x)dx$, where the function $y = f(x)$ is given as discrete set of points (x_i, y_i) , $i = 0, 1, 2, 3, \dots, n$, is given by

$$\int_a^b f(x)dx = \frac{h}{2}[y_0 + 2(y_1 + y_2 + \dots + y_{n-1}) + y_n]$$

Simpson's one-third rule

Simpson's one-third ($\frac{1}{3}$) rule to evaluate $\int_a^b f(x)dx$, where the function $y = f(x)$ is given as discrete set of points (x_i, y_i) , $i = 0, 1, 2, 3, \dots, n$, is given by

$$\int_a^b f(x)dx = \frac{h}{3} \left[\begin{array}{l} (y_0 + y_n) \\ +4(y_1 + y_3 + \dots + y_{n-1}) \\ +2(y_2 + y_4 + \dots + y_{n-2}) \end{array} \right].$$

Simpson's three-eighths rule

Simpson's three-eighths ($\frac{3}{8}$) rule to evaluate $\int_a^b f(x)dx$, where the function $y = f(x)$ is given as discrete set of points (x_i, y_i) , $i = 0, 1, 2, 3, \dots, n$, is given by

$$\int_a^b f(x)dx = \frac{3h}{8} \left[\begin{array}{l} (y_0 + y_n) \\ +3(y_1 + y_2 + y_4 + y_5 + \dots + y_{n-2} + y_{n-1}) \\ +2(y_3 + y_6 + \dots + y_{n-3}) \end{array} \right].$$

Example 4.2.1

Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ using

- (a) Trapezoidal rule taking $h = \frac{1}{5}$
- (b) Simpson's $\frac{1}{3}$ rule taking $h = \frac{1}{4}$
- (c) Simpson's $\frac{3}{8}$ rule taking $h = \frac{1}{6}$

(a). To solve $\int_0^1 \frac{1}{1+x^2} dx$ using trapezoidal rule. Taking

$$h = \frac{1}{5} = 0.2, n = \frac{b-a}{h} = \frac{1-0}{0.2} = 5$$

$\therefore$ Dividing the interval $(0, 1)$ into 5 equal parts for the function $f(x) = \frac{1}{1+x^2}$

x	0	0.2	0.4	0.6	0.8	1
$y = f(x)$	1	0.96	0.86	0.74	0.61	0.5

By trapezoidal rule,

$$\begin{aligned}\int_0^1 \frac{1}{1+x^2} dx &= \frac{h}{2} [y_0 + 2(y_1 + y_2 + y_3 + y_4) + y_5] \\ &= \frac{0.2}{2} [1 + 2(0.96 + 0.86 + 0.74 + 0.61) + 0.5] \\ &= 0.784.\end{aligned}$$

$\therefore \int_0^1 \frac{1}{1+x^2} dx = 0.784$ using trapezoidal rule.

(b). To solve $\int_0^1 \frac{1}{1+x^2} dx$ using Simpson's $\frac{1}{3}$ rule.

Taking

$$h = \frac{1}{4} = 0.25, n = \frac{b-a}{h} = \frac{1-0}{0.25} = 4$$

$\therefore$ Dividing the interval $(0, 1)$ into 4 equal parts for the function $f(x) = \frac{1}{1+x^2}$.

x	0	0.25	0.5	0.75	1
$y = f(x)$	1	0.94	0.8	0.64	0.5

By Simpson's $\frac{1}{3}$ rule,

$$\begin{aligned}\int_0^1 \frac{1}{1+x^2} dx &= \frac{h}{3} [(y_0 + y_4) + 4(y_1 + y_3) + 2(y_2)] \\ &= \frac{0.25}{3} [(1 + 0.5) + 4(0.94 + 0.64) + 2(0.8)] \\ &= 0.7850\end{aligned}$$

$\therefore \int_0^1 \frac{1}{1+x^2} dx = 0.7850$ using Simpson's $\frac{1}{3}$ rule.

(c). To solve $\int_0^1 \frac{1}{1+x^2} dx$ using Simpson's $\frac{3}{8}$ rule.

Taking

$$h = \frac{1}{6}, n = \frac{b-a}{h} = \frac{1-0}{\frac{1}{6}} = 6$$

$\therefore$ Dividing the interval into 6 equal parts for the function $f(x) = \frac{1}{1+x^2}$

x	0	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	1
$y = f(x)$	1	0.97	0.9	0.8	0.69	0.59	0.5

By Simpson's $\frac{3}{8}$ rule

$$\begin{aligned}\int_0^1 \frac{1}{1+x^2} dx &= \frac{3h}{8} [(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)] \\ &= \left(\frac{3}{8}\right) \left(\frac{1}{6}\right) [(1 + 0.5) + 3(0.97 + 0.9 + 0.69 + 0.59) + 2(0.8)] \\ &= 0.7844\end{aligned}$$

$\therefore \int_0^1 \frac{1}{1+x^2} dx = 0.7844$ using Simpson's $\frac{3}{8}$ rule.

Example 4.2.2

Evaluate $\int_0^{\frac{\pi}{2}} \sin x dx$ using

- a Trapezoidal rule
- b Simpson's $\frac{1}{3}$ rule
- c Simpson's $\frac{3}{8}$ rule, taking $n = 6$.

(a). Taking

$$n = 6, h = \frac{b - a}{n} = \frac{\frac{\pi}{2} - 0}{6} = \frac{\pi}{12}$$

∴ Dividing the interval $(0, \frac{\pi}{2})$ into 6 equal parts for the function $f(x) = \sin x$.

x	0	$\frac{\pi}{12}$	$\frac{2\pi}{12}$	$\frac{3\pi}{12}$	$\frac{4\pi}{6}$	$\frac{5\pi}{6}$	$\frac{\pi}{2}$
$y = f(x)$	0	0.2588	0.5	0.7071	0.866	0.9659	1

To solve $\int_0^{\frac{\pi}{2}} \sin x dx$ using trapezoidal rule

$$\begin{aligned} & \int_0^{\frac{\pi}{2}} \sin x dx \\ &= \frac{h}{2} [y_0 + 2(y_1 + y_2 + y_3 + y_4 + y_5) + y_6] \\ &= \left(\frac{\pi}{12}\right) \left(\frac{1}{2}\right) [0 + 2(0.2588 + 0.5 + 0.7071 + 0.866 + 0.9659) + 1] \end{aligned}$$

∴ $\int_0^{\frac{\pi}{2}} \sin x dx = 0.9943$ using trapezoidal rule.

(b). To solve $\int_0^{\frac{\pi}{2}} \sin x dx$ using Simpson's $\frac{1}{3}$ rule

$$\begin{aligned} & \int_0^{\frac{\pi}{2}} \sin x dx \\ &= \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)] \\ &= \left(\frac{\pi}{12}\right) \left(\frac{1}{3}\right) [(0 + 1) + 4(0.2588 + 0.7071 + 0.9659) + 2(0.5 + 0.866)] \end{aligned}$$

$\therefore \int_0^{\frac{\pi}{2}} \sin x dx = 1.000004$ using Simpson's $\frac{1}{3}$ rule.

(c). To solve $\int_0^{\frac{\pi}{2}} \sin x dx$ using Simpson's $\frac{3}{8}$ rule

$$\begin{aligned} & \int_0^{\frac{\pi}{2}} \sin x dx \\ &= \frac{3h}{8} [(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)] \\ &= \left(\frac{\pi}{12}\right) \left(\frac{3}{8}\right) [(0 + 1) + 3(0.2588 + 0.5 + 0.866 + 0.9659) + 2(0.7071)] \end{aligned}$$

$\therefore \int_0^{\frac{\pi}{2}} \sin x dx = 1.00004$ using Simpson's $\frac{3}{8}$ rule.

Gauss Quadrature 2-point formula

Gauss Quadrature 2-point formula for $I = \int_a^b f(x)dx$, is given by

$$I = \frac{b-a}{2} [w_1 f(x_1) + w_2 f(x_2)] \quad (6)$$

where,

$$x_1 = \frac{b-a}{2} z_1 + \frac{b+a}{2}$$

$$x_2 = \frac{b-a}{2} z_2 + \frac{b+a}{2}$$

$$w_1 = w_2 = 1$$

$$z_1 = \frac{-1}{\sqrt{3}}; z_2 = \frac{1}{\sqrt{3}}.$$

Gauss Quadrature 3-point formula

Gauss Quadrature 3-point formula for $I = \int_a^b f(x)dx$, is given by

$$I = \frac{b-a}{2} [w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3)] \quad (7)$$

where,

$$x_1 = \frac{b-a}{2} z_1 + \frac{b+a}{2}$$

$$x_2 = \frac{b-a}{2} z_2 + \frac{b+a}{2}$$

$$x_3 = \frac{b-a}{2} z_3 + \frac{b+a}{2}$$

$$w_1 = \frac{5}{9}, \quad w_2 = \frac{8}{9}, \quad w_3 = \frac{5}{9}$$

$$z_1 = -\sqrt{\frac{3}{5}}, \quad z_2 = 0, \quad z_3 = \sqrt{\frac{3}{5}}$$

Example 4.3.1

Use Gauss-Legendre two-point formula to evaluate

$$I = \int_{-2}^2 e^{\frac{-x}{2}} dx$$

Given data: $a = -2$, $b = 2$, and $f(x) = e^{\frac{-x}{2}}$

Gauss Quadrature 2-point formula:

$$I = \frac{b-a}{2} [w_1 f(x_1) + w_2 f(x_2)] \quad (8)$$

where,

$$x_1 = \frac{b-a}{2} z_1 + \frac{b+a}{2};$$

$$x_2 = \frac{b-a}{2} z_2 + \frac{b+a}{2}$$

$$w_1 = 1;$$

$$w_2 = 1$$

$$z_1 = \frac{-1}{\sqrt{3}};$$

$$z_2 = \frac{1}{\sqrt{3}}.$$

$$x_1 = \frac{b-a}{2}z_1 + \frac{b+a}{2} = \frac{2-(-2)}{2} \left(\frac{-1}{\sqrt{3}} \right) + \frac{2-2}{2} = \frac{-2}{\sqrt{3}}$$

$$f(x_1) = e^{\frac{-x_1}{2}} = e^{-\left(\frac{\left\lfloor \frac{-2}{\sqrt{3}} \right\rfloor}{2} \right)}$$

$$x_2 = \frac{b-a}{2}z_2 + \frac{b+a}{2} = \frac{2-(-2)}{2} \left(\frac{1}{\sqrt{3}} \right) + \frac{2-2}{2} = \frac{2}{\sqrt{3}}$$

$$f(x_2) = e^{\frac{-x_2}{2}} = e^{-\left(\frac{\left\lfloor \frac{2}{\sqrt{3}} \right\rfloor}{2} \right)}$$

$$\begin{aligned} I &= \frac{b-a}{2} [w_1 f(x_1) + w_2 f(x_2)] \\ &= \frac{2-(-2)}{2} \left[1 \times e^{-\left(\frac{\left\lfloor \frac{-2}{\sqrt{3}} \right\rfloor}{2} \right)} + 1 \times e^{-\left(\frac{\left\lfloor \frac{2}{\sqrt{3}} \right\rfloor}{2} \right)} \right] = 4.6854. \end{aligned}$$

Example 4.3.2

Use Gauss-Legendre three-point formula to evaluate

$$I = \int_2^4 (x^4 + 1)dx \quad (9)$$

Given data: $a = 2$, $b = 4$, and $f(x) = (x^4 + 1)$

Gauss Quadrature 3-point formula:

$$I = \frac{b-a}{2} [w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3)] \quad (10)$$

where,

$$x_1 = \frac{b-a}{2}z_1 + \frac{b+a}{2} \quad x_2 = \frac{b-a}{2}z_2 + \frac{b+a}{2} \quad x_3 = \frac{b-a}{2}z_3 + \frac{b+a}{2}$$

$$w_1 = \frac{5}{9}, \quad w_2 = \frac{8}{9}, \quad w_3 = \frac{5}{9}$$

$$z_1 = -\sqrt{\frac{3}{5}}, \quad z_2 = 0, \quad z_3 = \sqrt{\frac{3}{5}}$$



$$x_1 = \frac{b-a}{2}z_1 + \frac{b+a}{2}; x_2 = \frac{b-a}{2}z_2 + \frac{b+a}{2}; x_3 = \frac{b-a}{2}z_3 + \frac{b+a}{2}$$

$$x_1 = \frac{4-2}{2} \left(-\sqrt{\frac{3}{5}} \right) + \frac{4+2}{2} = 2.2254$$

$$x_2 = \frac{4-2}{2}(0) + \frac{4+2}{2} = 3$$

$$x_3 = \frac{4-2}{2} \left(\sqrt{\frac{3}{5}} \right) + \frac{4+2}{2} = 3.7746$$

$$f(x_1) = f(2.2254) = (2.2254^4 + 1) = 25.5263$$

$$f(x_2) = f(3) = (3^4 + 1) = 82$$

$$f(x_3) = f(3.7746) = (3.7746^4 + 1) = 203.9942$$

$$\begin{aligned}
 I &= \frac{b-a}{2} [w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3)] \\
 &= \left(\frac{4-2}{2}\right) \left[\left(\frac{5}{9}\right) (25.5263) + \left(\frac{8}{9}\right) (82) + \left(\frac{5}{9}\right) (203.9942) \right] \\
 &= 200.4014
 \end{aligned}$$

Example 4.3.3

Use Gauss-Legendre two-point formula to evaluate $I = \int_{-1}^1 e^x dx$

Given data: $a = -1$ $b = 1$ and $f(x) = e^x$ Using Gauss Quadrature 2-Point Formula:

$$I = \frac{b-a}{2} [w_1 f(x_1) + w_2 f(x_2)] \quad (11)$$

where,

$$w_1 = 1;$$

$$w_2 = 1$$

$$z_1 = \frac{-1}{\sqrt{3}};$$

$$z_2 = \frac{1}{\sqrt{3}}.$$

$$x_1 = \frac{b-a}{2} z_1 + \frac{b+a}{2} = \frac{1-(-1)}{2} \left(\frac{-1}{\sqrt{3}} \right) + \frac{1+1}{2} = \frac{-1}{\sqrt{3}}$$

$$x_2 = \frac{b-a}{2} z_2 + \frac{b+a}{2} = \frac{1-(-1)}{2} \left(\frac{1}{\sqrt{3}} \right) + \frac{1+1}{2} = \frac{1}{\sqrt{3}}$$

Gauss Quadrature 2-point formula:

$$\begin{aligned} I &= \frac{b-a}{2} [w_1 f(x_1) + w_2 f(x_2)] \\ &= \frac{1-(-1)}{2} \left[1 \times e^{-\left(\frac{-1}{\sqrt{3}}\right)} + 1 \times e^{\left(\frac{1}{\sqrt{3}}\right)} \right] = 2.3427. \end{aligned}$$

Example 4.3.4

Use Gauss-Legendre three-point formula to evaluate $I = \int_{-1}^1 e^x dx$

Given data: $a = -1$ $b = 1$ and $f(x) = e^x$ $x = \frac{b-a}{2}z + \frac{b+a}{2} = z$ Using Gauss Quadrature 3-Point Formula:

$$I = \frac{b-a}{2} [w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3)] \quad (12)$$

where,

$$\begin{aligned} x_1 &= \frac{b-a}{2}z_1 + \frac{b+a}{2} & x_2 &= \frac{b-a}{2}z_2 + \frac{b+a}{2} & x_3 &= \frac{b-a}{2}z_3 + \frac{b+a}{2} \\ w_1 &= \frac{5}{9}, & w_2 &= \frac{8}{9}, & w_3 &= \frac{5}{9} \\ z_1 &= -\sqrt{\frac{3}{5}}, & z_2 &= 0, & z_3 &= \sqrt{\frac{3}{5}} \end{aligned}$$

Gauss Quadrature 3-point formula:

$$x_1 = \frac{b-a}{2}z_1 + \frac{b+a}{2} = \frac{1-(-1)}{2} \left(-\sqrt{\frac{3}{5}} \right) + \frac{1+(-1)}{2} = -0.7745$$

$$f(x_1) = e^{(-0.7746)} = 0.4609$$

$$x_2 = \frac{b-a}{2}z_2 + \frac{b+a}{2} = \frac{1-(-1)}{2} (0) + \frac{1+(-1)}{2} = 0$$

$$f(x_2) = e^{(0)} = 1$$

$$x_3 = \frac{b-a}{2}z_3 + \frac{b+a}{2} = \frac{1-(-1)}{2} \left(\sqrt{\frac{3}{5}} \right) + \frac{1+(-1)}{2} = 0.7745$$

$$f(x_3) = e^{(0.7746)} = 2.1697$$

$$\begin{aligned} I &= \frac{b-a}{2} [w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3)] \\ &= \frac{1-(-1)}{2} \left[\frac{5}{9} \times (0.4609) + \frac{8}{9} \times (1) + \frac{5}{9} \times (2.1697) \right] = 2.3503. \end{aligned}$$